

Course Syllabus

[Communication](#)[Grading](#)[Assessments](#)[Assistance](#)[Policies](#)

Prerequisites

A grade of "P" or "C" or better in [CSCE 361](https://catalog.unl.edu/search/?P=CSCE%20361) (<https://catalog.unl.edu/search/?P=CSCE%20361>), [CSCE 361H](https://catalog.unl.edu/search/?P=CSCE%20361H) (<https://catalog.unl.edu/search/?P=CSCE%20361H>), [SOFT 261](https://catalog.unl.edu/search/?P=SOFT%20261) (<https://catalog.unl.edu/search/?P=SOFT%20261>), [SOFT 261H](https://catalog.unl.edu/search/?P=SOFT%20261H) (<https://catalog.unl.edu/search/?P=SOFT%20261H>), or [RAIK 284H](https://catalog.unl.edu/search/?P=RAIK%20284H) (<https://catalog.unl.edu/search/?P=RAIK%20284H>).

Meeting Times

11:00 AM–12:15 PM Central Time on Tuesdays and Thursdays

Meeting Location

Louise Pound Hall 27

Contact Information

[University of Nebraska-Lincoln](https://www.unl.edu/) (<https://www.unl.edu/>)

[College of Engineering](https://engineering.unl.edu/) (<https://engineering.unl.edu/>)

[School of Computing](https://computing.unl.edu/) (<https://computing.unl.edu/>)

Instructor: [Robert Dyer](https://go.unl.edu/rdyer) (<https://go.unl.edu/rdyer>)

Email: rdyer@unl.edu (<mailto:rdyer@unl.edu>)

Office: Avery 103E



(<https://go.unl.edu/rdyer>)

Office Hours: open-door policy (reach out via email)

or

Mondays 11:30 AM–12:30 PM

Thursdays 2:00–3:00 PM

Fridays 3:00 PM–4:00 PM

Teaching Assistant: Idriss Abdelmadjid

Email: iabdelmadjid2@huskers.unl.edu

[\(mailto:iabdelmadjid2@huskers.unl.edu\)](mailto:iabdelmadjid2@huskers.unl.edu)

Office: SRC

Office Hours: Tuesdays 12:30–1:30 PM

Thursdays 12:30–1:30 PM

Course Description

From the official course description:

In-depth coverage of problems related to software quality, and approaches for addressing them. Topics include testing techniques, dynamic and static program analysis techniques, and other approaches for verifying software qualities. Tool support for performing testing, verification, and analysis will also be studied.

Course Objectives

After completing this course, students should have:

1. Mastery: "basic" black-box, white-box, model-based, search-based, and fault-based testing techniques applied at various levels of testing (unit, integration, system).
2. Familiarity: regression testing and techniques for performing it.

3. Exposure: issues for testing in special domains such as testing ML-generated software (e.g., Deep Neural Networks); advanced approaches for verification, such as symbolic analysis, abstract interpretation, and automatic test case generation.

Communication

Course materials will be posted on Canvas, and course announcements will also be posted on Canvas. Students should configure their Canvas notification settings so that they receive these announcements in a timely manner. It is School of Computing policy that all students in SOC courses are expected to regularly check their email for notifications so that they do not miss important announcements.

The primary method for contacting the instructor outside of class is email.

The instructor has an open-door policy for office hours and can be reached via email when not otherwise booked; the instructor also has regular office hours, which are held in Avery 103E.

Additionally, the **[SOC Student Resource Center \(SRC\)](https://computing.unl.edu/current-undergraduate#SRC)** (<https://computing.unl.edu/current-undergraduate#SRC>) is staffed by student tutors who are available to help you with this course or with issues such as problems logging in to SOC systems, problems printing, installing an application, etc. The SRC also provides a study space in Avery 12 that is open to all software engineering majors.

The School of Computing also maintains **[an anonymous suggestion box](https://computing.unl.edu/anonymous-feedback-form/)** (<https://computing.unl.edu/anonymous-feedback-form/>) that you may use to voice your concerns about any problems in the course or the school, if you do not wish to be identified. These concerns can be submitted at any time.

If in-person classes are canceled, you will be notified of the instructional continuity plan for this class on Piazza.

Grading

Assessment (grading) will be based on homework, programming assignments, exams, and a course project with the following weights:

Category	Number	Percent Each	Total
Assignments	4	5%	20%
Programs	4	5%	20%

Project	1	30%	30%
Exam 1	1	15%	15%
Exam 2	1	15%	15%
Total			100%

Course scores will be rounded to the nearest percent, and letter grades will be assigned according to the following rubrics:

SOFT/CSCE 467						CSCE 867					
A+	$\geq 97\%$	A	$\geq 93\%$	A-	$\geq 90\%$	A+	$\geq 99\%$	A	$\geq 95\%$	A-	$\geq 92\%$
B+	$\geq 87\%$	B	$\geq 83\%$	B-	$\geq 80\%$	B+	$\geq 89\%$	B	$\geq 85\%$	B-	$\geq 82\%$
C+	$\geq 77\%$	C	$\geq 73\%$	C-	$\geq 70\%$	C+	$\geq 79\%$	C	$\geq 75\%$	C-	$\geq 72\%$
D+	$\geq 67\%$	D	$\geq 63\%$	D-	$\geq 60\%$	D+	$\geq 69\%$	D	$\geq 65\%$	D-	$\geq 62\%$
F	$< 60\%$					F	$< 62\%$				

The scales may be adjusted if the instructor deems it necessary, based on the final course grades.

The instructor/TA will make every effort to grade and return submitted material within two academic weeks from the due date. If you have questions about your grade or believe that points were deducted unfairly, you must address the issue with the TA within one week after the graded assignment is returned to you. If the TA is unable to address the issue, please contact the instructor. We make every attempt to assign grades consistently on each assignment; we can do this only if we grade everyone's work at the same time.

Late Work

In general, there will be no make-up exams or late work accepted. Exceptions may be made in certain circumstances, such as health or emergency, but you must make every effort to get **prior** permission. Documentation may also be required. Assignments and programs will be submitted on Gradescope, and Gradescope locks the assignment at the due date. Any late submission, even 1 second late, will be rejected by Gradescope, so please be sure to submit early. Gradescope allows multiple submissions up to the due date, and the last submission is the one that is graded.

Assessments

Readings

This course is organized into weekly modules, and each module has a reading associated with it. The textbook is online and free. Not every set of required reading represents an equal amount of work. Some modules have more reading than others.

Assignments

The assignments are your way of seeing where the gaps in your knowledge are. Assignments are required and graded. Assignments are to be completed outside of class, and solutions will be provided and discussed in a lecture, after the due date.

Programs

There will be 4 programming assignments throughout the semester. These are to be completed individually - no collaboration is allowed. Programs must be written in Python and run on Gradescope without installing additional packages. Every program must include a README.md file explaining how to run it, various design decisions that were made, etc. The instructor may provide a template/helper code for the assignments, but you are not required to use it.

Any assignment that has a compile error and/or does not run on Gradescope will be given a score of 0 on any automated test rubrics.

Project

There is a semester-long, team-based (two-person) project. A project proposal will be due earlier in the semester, and you must receive approval from the instructor before continuing with your project. A final report and presentation will be required. Given the size of the course, presentations will be recorded. Teams will be randomly chosen to play their recording during a lecture and answer questions (not all teams will present to the class).

Exams

There will be one midterm exam and a non-comprehensive second exam due during finals week. More details will be announced closer

to the exam dates.

Sources for Help and Assistance

You are ultimately responsible for your success in this course. If you have questions on material covered or assigned in class, it is up to you to seek out assistance from the course instructor. Staff in the [SOC Student Resource Center \(https://computing.unl.edu/current-undergraduate#SRC\)](https://computing.unl.edu/current-undergraduate#SRC) may also be able to assist you with general questions. The School of Computing also maintains [a Frequently Asked Questions page \(https://computing.unl.edu/faq\)](https://computing.unl.edu/faq).

Writing Center

The Writing Center can provide you with meaningful support as you write for this class, other classes, or for non-academic purposes. You are welcome to bring in everything from lab reports, presentations, and research papers to cover letters, application essays, and graduate theses and dissertations. Trained undergraduate and graduate peer consultants are available to talk with you as you plan, draft, and revise your writing, both in person and online. You don't need to have anything written yet: Writing Center Consultants are happy to sit down and brainstorm with you as soon as you get an assignment, as well as help you organize your ideas or polish a final draft.

There are **three** ways you can connect with a Consultant: **In Person**, **Online** (a real-time, video conversation), and **eTutoring** (email feedback). **In-person Writing Center appointments are held in Andrews Hall 102.** Writers may choose between **25-** and **50-minute** appointments. To learn more about online options and view video tutorials, visit <https://writing.unl.edu/online-writing-center-services/> (<https://writing.unl.edu/online-writing-center-services/>).

The Writing Center also offers [accountability appointments \(https://writing.unl.edu/accountability-appointments/\)](https://writing.unl.edu/accountability-appointments/), in which you can sit beside a Writing Center consultant and talk as much or as little as you like about your writing. You can indicate that this is what you're looking for when you sign up for an appointment.

Sign up anytime by visiting unl.mywconline.com  (<https://unl.mywconline.com/>). For more information about the Writing Center, please visit <https://writing.unl.edu/> (<https://writing.unl.edu/>).

Dead Week Policy

In compliance with UNL's 15th Week Policy (see the [university-wide course policies \(https://executivevc.unl.edu/academic-excellence/teaching-resources/course-policies\)](https://executivevc.unl.edu/academic-excellence/teaching-resources/course-policies)), be aware that Program 4, project presentations, peer evals, and project reports will be due during the final week of classes. Note also that all assignments, homework, projects, etc., will have a strict final due date during the final week of classes.

Computer Policy

The computer policy for this course is the same as [the computer policy for the software engineering \(SE\) major \(https://computing.unl.edu/software-engineering-computer-policy\)](https://computing.unl.edu/software-engineering-computer-policy). If you are not a SE major and you have concerns regarding this policy, please see the instructor as soon as possible to make alternative arrangements.

Academic Integrity

The School of Computing has an [Academic Integrity Policy \(https://computing.unl.edu/academic-integrity-policy\)](https://computing.unl.edu/academic-integrity-policy), which all students enrolled in any software engineering course are bound by. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case-by-case basis and may result in a failing assignment or a failing grade for the course itself.

The use of A.I. tools such as ChatGPT and GitHub CoPilot is allowed in this course under some circumstances. A.I. should not replace your individual effort or original work but rather should be used as a supplemental resource to support your own analysis, critical thinking, and problem-solving.

For homework assignments and programming assignments, you must clearly indicate any use of A.I. tools and provide appropriate citations or references for any A.I.-generated content or results produced. This should include full documentation of exactly how the tool was used, including how you prompted it and if you revised the generated response.

For exams, the specific use of generative A.I. tools will be clarified.

Any misuse or violation of the policy, including unauthorized or excessive use of A.I., will be considered a breach of academic integrity and subject to disciplinary actions as per the institution's policies and procedures on academic misconduct.

University-wide Policies and Resources

Please also see the additional, [university-wide course policies and resources \(http://go.unl.edu/coursepolicies\)](http://go.unl.edu/coursepolicies).

Course Summary:

Date	Details	Due
	 Extra Credit: Hour of Code (https://canvas.unl.edu/courses/201132/assignments/2029000)	
	 Extra Credit: Research Study (https://canvas.unl.edu/courses/201132/assignments/2029001)	